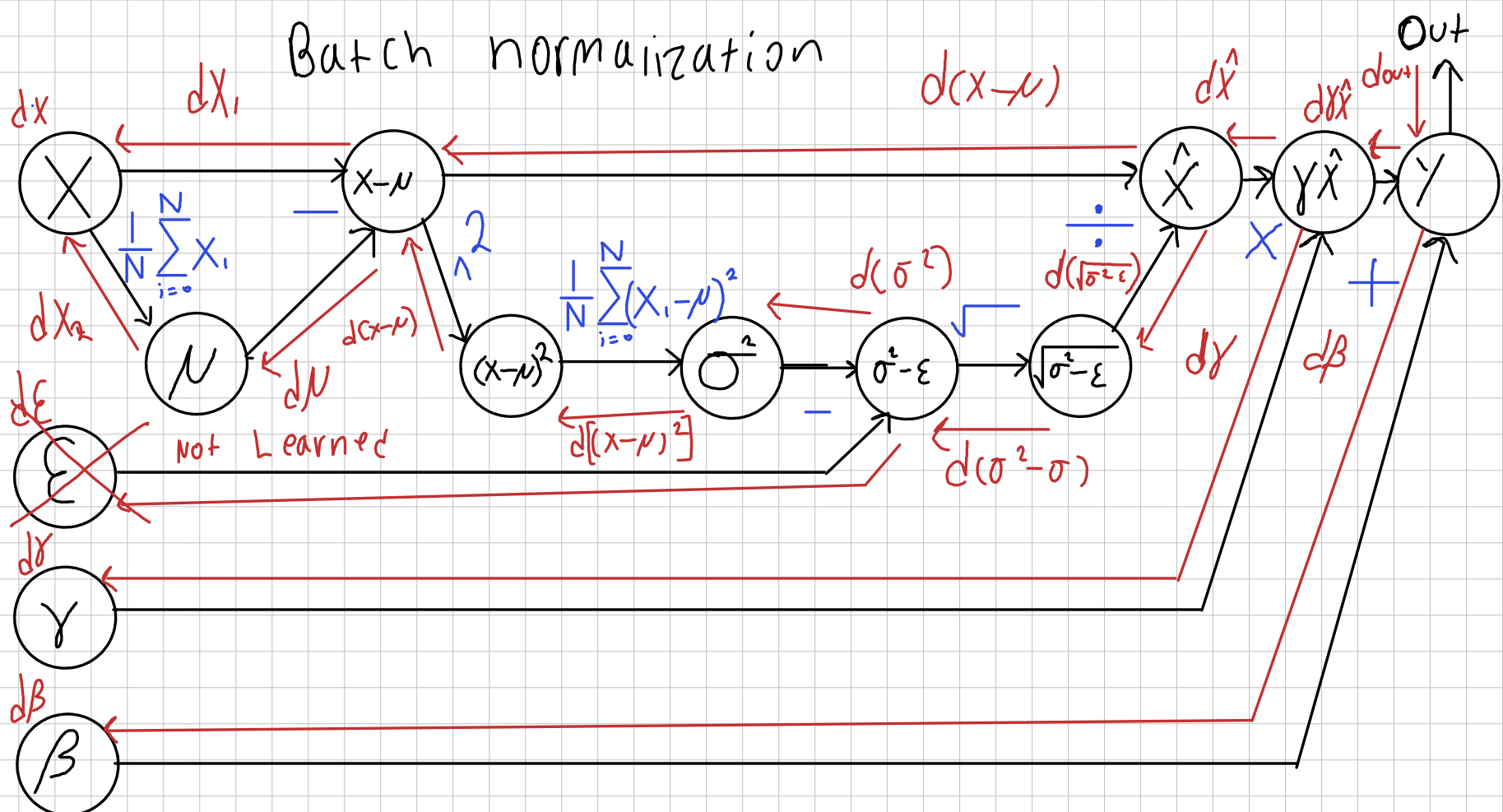




$$\text{dout} \in \mathbb{R}^{N \times D} \quad \beta \in \mathbb{R}^D \quad \gamma \hat{X} \in \mathbb{R}^{N \times D} \quad \hat{X} \in \mathbb{R}^{N \times D}$$

$$\gamma \in \mathbb{R}^D \quad \sqrt{\sigma^2 + \epsilon} \in \mathbb{R}^D \quad (\sigma^2 - \epsilon) \in \mathbb{R}^D \quad \sigma^2 \in \mathbb{R}^D$$

$$(x - \mu)^2 \in \mathbb{R}^{N \times D} \quad (x - \mu) \in \mathbb{R}^{N \times D} \quad \mu \in \mathbb{R}^D \quad X \in \mathbb{R}^{N \times D}$$

$$d\hat{X} = \frac{dY}{d\hat{X}} \cdot \text{dout} = \frac{d(\gamma \hat{X} + \beta)}{d\hat{X}} \cdot \text{dout} = \text{dout}$$

$$d\beta = \frac{dY}{d\beta} \cdot \text{dout} = \sum \text{dout} \quad \text{because } \beta \text{ is a constant but is added in a per element manner}$$

$$d\hat{X} = \frac{d\gamma \hat{X}}{d\hat{X}} \cdot d\gamma = \frac{d\gamma \hat{X}}{d\hat{X}} \cdot d(\gamma \hat{X} + \beta) = \gamma \cdot d\hat{X} = \gamma \text{dout}$$

$$d\gamma = \frac{d\gamma \hat{X}}{d\gamma} \cdot d\gamma = \hat{X} d\hat{X} = \hat{X} \text{dout}$$

$$d(\sqrt{\sigma^2 + \epsilon}) = d\hat{X} \cdot \frac{d\left(\frac{(x - \mu)^2}{(\sigma^2 - \epsilon)^{1/2}}\right)}{d\hat{X}} = d\hat{X} \cdot \left(-\frac{x - \mu}{(\sigma^2 - \epsilon)^{3/2}}\right)$$

$$d(x - \mu) = d\hat{X} \cdot \frac{1}{\sqrt{\sigma^2 + \epsilon}} \quad (\sigma^2 + \epsilon)^{1/2} : 2 \cdot \frac{1}{2}$$

$$d\sigma^2 = \frac{d(\sqrt{\sigma^2 + \epsilon})}{d\sigma^2} = d(\sqrt{\sigma^2 + \epsilon}) \cdot \frac{1}{2\sqrt{\sigma^2 + \epsilon}}$$

$$d[(x - \mu)^2] = \frac{d\sigma^2}{d[(x - \mu)^2]} = d\sigma^2 \cdot \frac{d\left(\frac{1}{N} \sum (x_i - \mu)^2\right)}{d[(x - \mu)^2]} = d\sigma^2 \frac{1}{N} \begin{bmatrix} 1 \\ 1 \end{bmatrix}^{N \times D}$$

$$d(x - \mu)_1 = \frac{d[(x - \mu)^2]}{d(x - \mu)} = \frac{d[(x - \mu)^2]}{d(x - \mu)} = 2(x - \mu) d[(x - \mu)^2]$$

$$d\mu = \frac{d(x - \mu)}{d\mu} = d(x - \mu) \cdot -1 = -1 \cdot (d(x - \mu)_1 + d(x - \mu)_2)$$

$$dx_1 = \frac{d(x - \mu)}{dx_1} = d(x - \mu)_1 + d(x - \mu)_2$$

$$dx_2 = \frac{1}{N} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot d\mu$$

$$dX = dx_1 + dx_2$$

$$2.0 \quad o[n, f, i, j] = \sum_{s=0}^{f-1} \sum_{h=0}^{h-1} \sum_{w=0}^{W-1} X[n, :, i \cdot s + h, j \cdot s + w] \cdot W[f, :, h, w] + b[f]$$

• filters applied to all channels therefore : in c index
 • Per Pixel Gradient

$$2.1 \quad \frac{\partial L[n, f, h, w]}{\partial W[f, :, h, w]} = \frac{\partial L_{\text{pixel}}}{\partial o_{\text{pixel}}} \cdot \frac{\partial o_{\text{pixel}}}{\partial W_{\text{pixel}}}$$

$$= \sum_{s=0}^{f-1} \sum_{h=0}^{h-1} \sum_{w=0}^{W-1} X[n, :, i \cdot s + h, j \cdot s + w] \cdot \text{dout}[n, f, h, w]$$

$$2.2 \quad \frac{\partial L[n, f, h, w]}{\partial W[f, :, h, w]} = \frac{\partial L_{\text{pixel}}}{\partial o_{\text{pixel}}} \cdot \frac{\partial o_{\text{pixel}}}{\partial X_{\text{pixel}}}$$

$$= \sum_{s=0}^{f-1} \sum_{h=0}^{h-1} \sum_{w=0}^{W-1} W[f, :, h, w] \cdot \text{dout}[n, f, h, w]$$

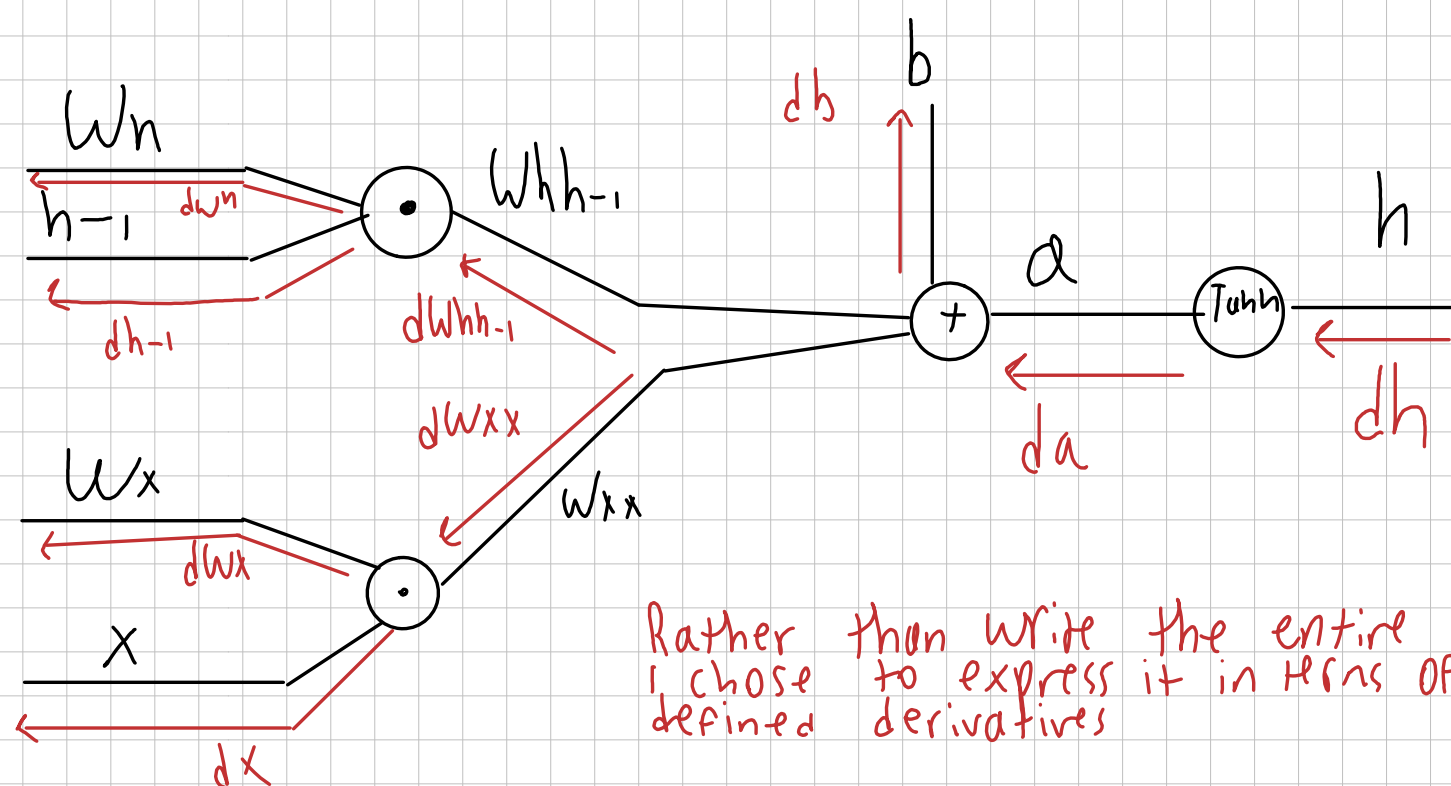
$$2.3 \quad \frac{\partial L_{\text{pixel}}}{\partial b[f]} = \frac{\partial L_{\text{pixel}}}{\partial o_{\text{pixel}}} \cdot \frac{\partial o_{\text{pixel}}}{\partial X_{\text{pixel}}}$$

$$= \sum_{s=0}^{f-1} \sum_{h=0}^{h-1} \sum_{w=0}^{W-1} 1 \cdot \text{dout}[n, f, h, w]$$

We perform this accumulation because, despite being a single scalar per filter, the bias is applied to each element

Max pooling does not involve arithmetic in the same manner as the other problems here. Therefore backprop simply means applying the inputted gradient to the pixel that is used in the max pooling

RNN COMP Graph $h = \tanh(x \cdot w_x + h_{-1} \cdot w_h + b)$



$$4.1 \quad \frac{da}{dh} = \frac{d \tanh(a)}{da} dh = (1 - \tanh(a)^2) dh = (1 - h^2) dh$$

$$4.2 \quad \frac{db}{db} = \frac{d(W_h h_{-1} + W_x x + b)}{db} \cdot da = \sum da$$

b appears for each element

so we need to sum

$$4.3 \quad \frac{dW_h h_{-1}}{d(W_h h_{-1})} = \frac{d(W_h h_{-1} + W_x x + b)}{d(W_h h_{-1})} da = 1 \cdot da$$

Useless computation
↓

$$4.4 \quad \frac{dW_h}{d(W_h)} = \frac{d(W_h h_{-1} + W_x x + b)}{d(W_h)} da = da \cdot (h_{t-1})^T$$

$$4.5 \quad dh_{-1} = W_h^T da$$

$$4.6 \quad \frac{dW_x}{d(W_x)} = \frac{d(W_h h_{-1} + W_x x + b)}{d(W_x)} da = da x^T$$

$$4.7 \quad dx = W_x^T da$$

The diagram illustrates the internal structure of an LSTM cell and its connections to the previous time step. The cell's state is represented by c_t (cell state) and h_t (hidden state). The inputs are x_t and the previous hidden state h_{t-1} . The cell's internal operations involve the forget gate (f), input gate (i), and output gate (o), each using a sigmoid (σ) or tanh activation function. The cell state is updated as $c_t = f \odot c_{t-1} + i \odot g$, where g is the candidate cell state. The hidden state is updated as $h_t = \tanh(c_t) \odot o$. The diagram includes numerous handwritten red arrows and labels indicating the flow of gradients during backpropagation through time (BPTT). Key labels include dc_{t-1} , $dc_{t-total}$, dc_t , dh_t , dh_{t-1} , dx_t , dw_{hx} , dw_{hh} , dw_{wh} , db , df , di , dg , do , da_t , dai , dag , dah , $dact$, and $dactanh$. A note at the bottom right states: "Same as last + in terms of previous derivatives".

$$h_t = \tanh(c_t)$$

$$5.1 \frac{dG(t)}{dt} = \frac{d \tanh(Ct)}{dt} \cdot d \tanh(Ct) = d \tanh(Ct) \cdot (1 - \tanh(Ct)^2)$$

$$5.3 \quad dO = dht \cdot \tanh(ct)$$

$$5.5 \quad dg = dCt_{total} \cdot i$$

$$5.7 \quad d_i = d(t_{\text{total}}) \cdot g$$

$$5.8 \quad da_i = d_i \cdot d\sigma_i = d_i \cdot \dot{I} \cdot (1 - \dot{I})$$

$$5.2 \quad df = dcf_total \cdot i$$

$$5.10 \quad da_f = df \cdot \sigma(f) = df \cdot f \cdot (1-f)$$

5.11 $da = [da_i; da f'; da o'; da g]$

$$5.12 \quad dx = \frac{d(WxX)}{dx} da = da Wx^T$$

S.13 $dw_x = \frac{d(W \times X)}{dw_x} \cdot da = X^T \cdot da$

$$\text{S.o. } d\mathbf{h}_{t-1} = \frac{d(\mathbf{W}\mathbf{h}_{t-1})}{d} \cdot d\mathbf{a} = d\mathbf{a}\mathbf{W}_h^T$$

$$\text{s.15 } dW_h = \frac{d(W_{h,t-1})}{dw} \cdot da = h_{t-1}^T \cdot da$$

5.15 $d_b = \sum d_a$ because b is affixed across each element